Mathematics - Graduate Level Assessment 1 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. If f(z) is an analytic function that maps the entire finite complex plane into the real axis, then the imaginary axis must be mapped onto
   (a) a circle
   (b) an infinite interval
   (c) the negative real axis
   (d) the positive real axis
   (e) a ray

2. The original price of a new bicycle is $138.00. If the bicycle is marked down 15%, what is the new price?
   (a) $125.00
   (b) $20.70
   (c) $130.00
   (d) $133.00
   (e) $115.00

3. Consider the initial value problem $y'+\frac{1}{4}y = 3+2\cos 2t, \quad y(0) = 0 Determinethevalueof t forwhichthe$
   (a) 11.234567
   (b) 10.065778
   (c) 6.789012
   (d) 7.654321
   (e) 9.876543

4. Write the ratio of 96 runners to 216 swimmers in simplest form.
   (a) 16 over 36
   (b) 9 over 4
   (c) 8 over 18
   (d) 48 over 108
   (e) 24 over 54

5. In the group G = {2, 4, 6, 8) under multiplication modulo 10, the identity element is
   (a) 9
   (b) 7

(c) 8

(d) 1

(e) 4

True / False (10 marks)

Answer True or False

6. True or False: Including 12 people in a group guarantees that at least two will have birthdays in the same month.

7. True or False: The cumulative distribution function of the standard Gaussian distribution is log-concave.

8. True or False: The double integral of the function f(x,y)=x*y^3 exp$^{\{}$x^2+y^2} over the region I=[0,1]×[0,1] equals 0.5234.

9. True or False: When selecting 3 toppings from 8 available options, there are 72 different combinations possible.

10. True or False: The number that satisfies the equation is 17, as it fulfills the condition of the problem statement.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. For a certain natural number n, n^2 gives a remainder of 4 when divided by 5, and n^3 gives a remainder of 2 when divided by 5. What remainder does n give when divided by 5?

12. Given the equation $|x - 4| - 10 = 2$, analyze the steps to solve for x and discuss the significance of the product of all possible values of x.

End of Paper